

SUMMER OF SCIENCE 2026

CS03 – Artificial Intelligence and Machine Learning

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Week-wise Breakdown

WEEK 1 (24–30 May): Introduction to Machine Learning & Python Fundamentals

- Introduction to Artificial Intelligence, Machine Learning, and Deep Learning
- Types of Machine Learning and real-world applications
- Python programming fundamentals and data structures
- Introduction to NumPy, Pandas, Matplotlib, and Seaborn
- Basic data analysis and visualization exercises

WEEK 2 (31 May–6 June): Data Preprocessing & Exploratory Data Analysis

- Data collection and importing datasets
- Handling missing values and noisy data
- Data standardization and feature encoding
- Train-test splitting and preprocessing pipelines
- Exploratory Data Analysis (EDA) and visualization techniques
- Working with numerical and textual datasets

WEEK 3 (7–13 June): Mathematics for Machine Learning & Basic ML Models

- Linear Algebra, Statistics, and Probability for Machine Learning
- Supervised and Unsupervised Learning concepts
- Linear Regression and Logistic Regression
- Gradient Descent, Loss Functions, and Model Evaluation
- Introduction to model parameters and hyperparameters
- Implementing basic regression and classification models

WEEK 4 (14–20 June): Advanced Supervised Learning & Midterm Report Preparation

- Support Vector Machines (SVMs) and kernel methods
- Decision Trees and ensemble learning concepts
- Overfitting, Underfitting, and Bias-Variance Trade-off
- Cross-validation and Hyperparameter Tuning
- Model selection and performance evaluation metrics
- Midterm Report Preparation and documentation of progress

WEEK 5 (21–27 June): Applied Machine Learning & Case Studies

- Building end-to-end Machine Learning workflows
- Feature engineering and preprocessing optimization
- Classification and prediction use cases on real-world datasets
- Clustering and introductory unsupervised learning techniques
- Comparative analysis of multiple Machine Learning models
- Mini-projects based on classification and prediction problems

WEEK 6 (28 June–4 July): Deep Learning & Reinforcement Learning Basics

- Introduction to Neural Networks and Deep Learning concepts
- Image data processing and introductory neural network implementations
- Neural network based classification tasks
- Introduction to Reinforcement Learning concepts and applications
- Understanding practical applications of Deep Learning and RL
- Exploring basic Deep Learning workflows and implementations

WEEK 7 (5–11 July): Final Project Development – Spaceship Titanic Dataset

- Understanding the project dataset and performing Exploratory Data Analysis (EDA)
- Data preprocessing, feature engineering, and feature selection techniques
- Training, evaluating, and comparing multiple classification models
- Cross-validation, error analysis, and Kaggle submission workflow
- Streamlit installation and Machine Learning model deployment
- Building interactive Machine Learning web applications and final model evaluation

WEEK 8 (12–18 July): Finalisation & End-term Submission

- Final Kaggle submissions and result comparison
- Final model refinement, deployment polishing, and testing
- Preparing visualizations and project summaries
- Finalizing GitHub repository and documentation
- End-term Report Submission

Resources

1. **Machine Learning by Stanford University – Andrew Ng (Coursera)**
 - [Andrew Ng Machine Learning Course](#)
2. **StatQuest with Josh Starmer**
 - [StatQuest Machine Learning Playlist](#)
3. **Primary Machine Learning Playlist Followed During SoS**
 - [Machine Learning Full Course Playlist](#)
4. **Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow Book by Aurélien Geron**
5. **GeeksforGeeks – Machine Learning**
 - [GeeksforGeeks Machine Learning](#)
6. **Python & Libraries Tutorials**
 - [Basic Python Tutorial](#)
 - [NumPy Tutorial](#)
 - [Pandas Tutorial](#)
 - [Matplotlib Tutorial](#)
 - [Seaborn Tutorial](#)